

FIREWISE COMMUNITIES/USA® RECOGNITION PROGRAM

COMMUNITY ASSESSMENT TEMPLATE

Homestead Valley Firewise Community

10/28/15 (revision3)



This text is prepared for the use of individuals conducting community assessments as a part of the Firewise Communities/USA® Recognition Program. It is meant for insertion in those sections of the document cited.

1) Introduction:

The Firewise Communities/USA program is designed to provide an effective management approach for preserving wildland living aesthetics. The program can be tailored for adoption by any community and/or neighborhood association that is committed to ensuring its citizens maximum protection from wildland fire. The following community assessment is intended as a resource to be used by the Homestead Valley residents for creating a wildfire safety action plan. The plan developed from the information in this assessment should be implemented in a collaborative manner, and updated and modified as needed.

This assessment was conducted by Captain Cary Gloeckner from Southern Marin Fire Department, Station #4 (309 Poplar Avenue, Mill Valley, CA 94941).

2) Definition of the Home Ignition Zone:

Homestead Valley is located in a wildfire environment. Wildfires will happen--exclusion is not a choice. The variables in a fire scenario are when the fire will occur, and where. This assessment addresses the wildfire-related characteristics of Homestead Valley. It examines the area's exposure to wildfire as it relates to ignition potential. The assessment does not focus on specific homes, but examines the community as a whole.

A house burns because of its interrelationship with everything in its surrounding home ignition zone---the house and its immediate surroundings. To avoid a home ignition, a homeowner must eliminate the wildfire's potential relationship with his/her house. This can be accomplished by interrupting the natural path a fire takes. Changing a fire's path by clearing a home ignition zone is an easy-to-accomplish task that can result in avoiding home loss. To accomplish this, flammable items such as dead vegetation must be removed from the area immediately around the structure to prevent flames from contacting it. Also, reducing the volume of live vegetation will affect the intensity of the wildfire as it enters the home ignition zone.

Included in this assessment are observations made while visiting Homestead Valley. The assessment addresses the ease with which home ignitions can occur under severe wildfire conditions and how these ignitions might be avoided within the home ignition zones of affected residents. Homestead Valley residents can reduce their risk of destruction during a wildfire by taking actions within their home ignition zones. This zone principally determines the potential for home ignitions during a wildland fire; it includes a house and its immediate surroundings within 100 to 150 feet.

The result of the assessment is that wildfire behavior will be dominated by the residential characteristics of this area. The good news is that by addressing community vulnerabilities, residents will be able to substantially reduce their exposure to loss. Relatively small investments of time and effort will reap great rewards in wildfire safety.

3) Description of the severe case wildland fire characteristics that could threaten Homestead Valley:

Fire intensity and spread rate depend on the fuel type and condition (live/dead), the weather conditions prior and during ignition, and the topography. Generally the following relationships hold between the fire behavior and the fuel, weather and topography.

- Fine fuels ignite more easily and spread faster with higher intensities than coarser fuels. For a given fuel, the more there is and the more continuous it is, the faster the fire spreads and the higher the intensities. Fine fuels take a shorter time to burn out than coarser fuels.
- The weather conditions affect the moisture content of the dead and live vegetative fuels. Dead fine fuel moisture content is highly dependent on the relative humidity and the degree of sun exposure. The lower the relative humidity and the greater the sun exposure, the lower will be the fuel moisture content. Lower fuel moistures produce higher spread rates and fire intensities.
- Wind speed significantly influences the rate of fire spread and fire intensity. The higher the wind speed, the greater the spread rate and intensity.
- Topography influences fire behavior principally by the steepness of the slope. However, the configuration of the terrain such as narrow draws, saddles and so forth can influence fire spread and intensity. In general, the steeper the slope, the higher the uphill fire spread and intensity.

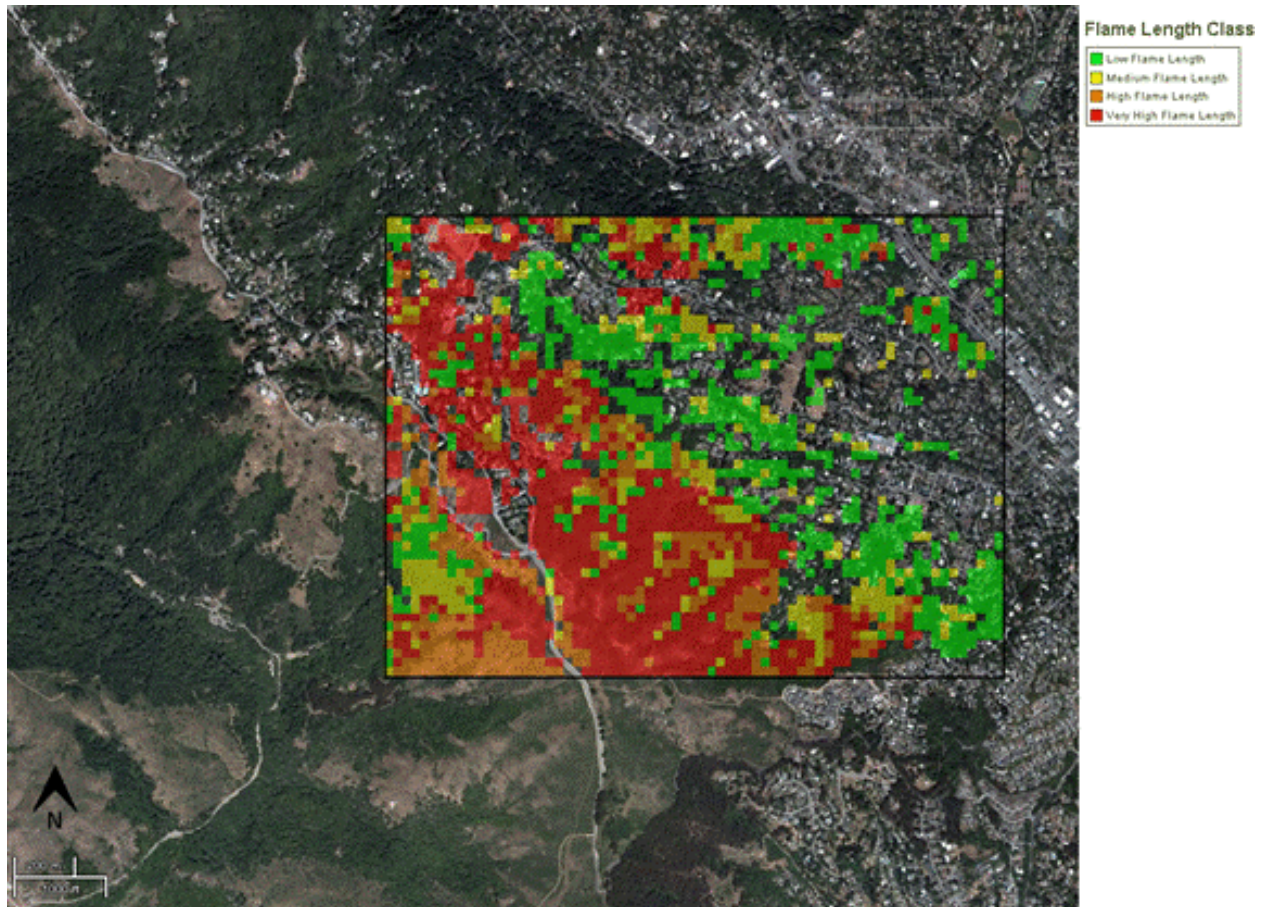
TWO MAIN WILDFIRE SCENARIOS CONSIDERED

There are two main scenarios that we are highly concerned about defending against. The first and primary concern is a conflagration type event. We are very concerned about the three to five day weather event in October where we commonly experience hot dry weather. Low relative humidity between 10 and 20% over this time span coupled with temperatures between 88 and 100F for this three to five day lead up are its most critical determinant factors when predicting this kind of event. It is very common for winds at this time to flow offshore either N, NE or East winds sustained at 15 to 20 mph gusting to 40 mph.

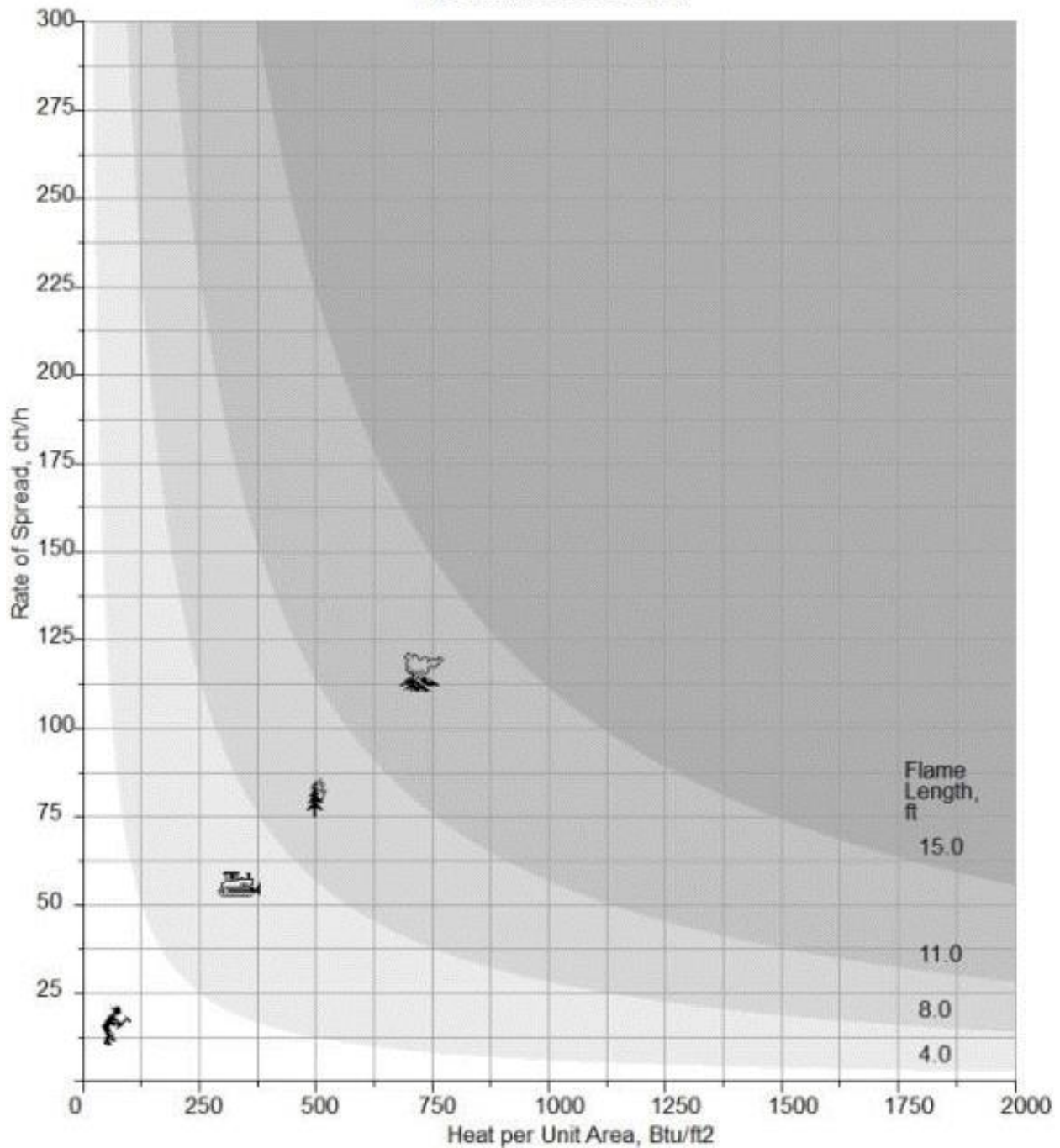
This allows the forward progress of the fire to make significant gains in total acreage consumed, fire line perimeter and number of homes consumed. The more homes that become involved with fire, the larger the ember shower cast will become and that coupled with smoke occlusion to aircraft assets will make containment nearly impossible. This kind scenario assumes that once the fire transitions from small to large, homes and people will essentially be on their own to flee the ensuing flames.

IFTDSS worst case fire behavior computer modeling produced the following maps illustrating our worst case areas. The Fire District would hypothesize a fire occurring under these conditions, given the three to five day lead up, that would start as either a small vegetation fire, car fire or structure fire that serves to develop a large ember cast shower over the near area. Once three to five structures are impacted with active fire, our chances for successfully containing the fire go down dramatically. This figure is derived from assuming that one attic/roof fire would generally commit three engines, a rescue and one Battalion Chief to properly control. Three simultaneous events during a wildfire dispatch would consume all available alarm resources county wide prompting the use of strike teams from out of county to fill the balance of the firefighting forces. The time frame for the first strike team from Sonoma County to arrive would be anywhere from 60 to 90 minutes from time of order to on the ground fighting fire.

Offshore Flow Flame Length Map – OCTOBER 2015



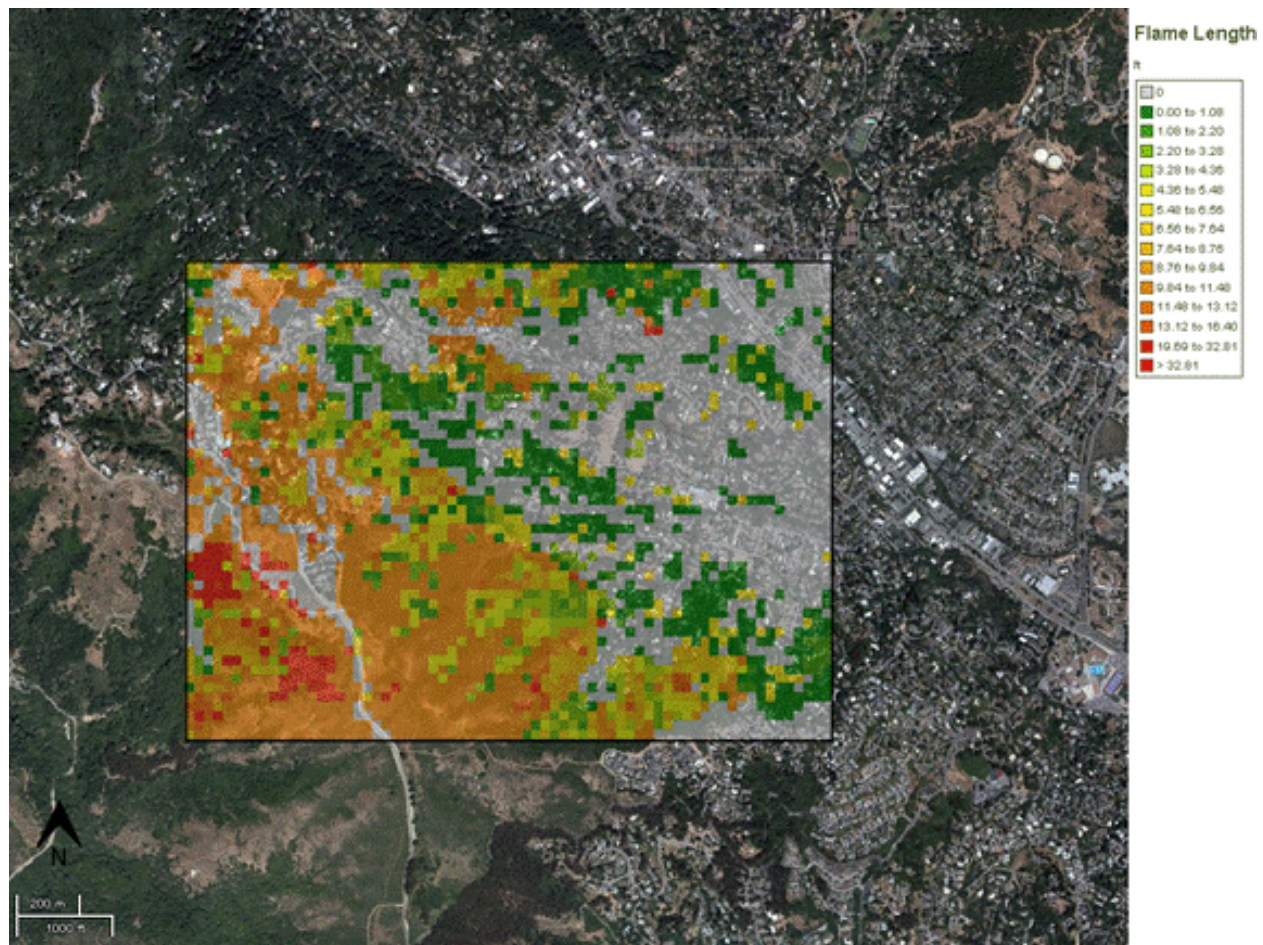
SURFACE FIRE BEHAVIOR Fire Characteristics Chart



As you can see, areas of the west and southwest portions of Homestead Valley have the highest flame lengths under the offshore flow computer modeling which concerns the Fire District the most given that when reflecting upon the Fire Behavior Characteristics Chart or “Hauling Chart”, firefighters would be put in extreme danger while attempting to conduct a frontal assault on this type of fire or when placed out front of this kind of fire for structure protection without excellent defensible space and or fuel treatments near the developments. See hauling chart below showing the capacity to fight fire given a person and other fire equipment.

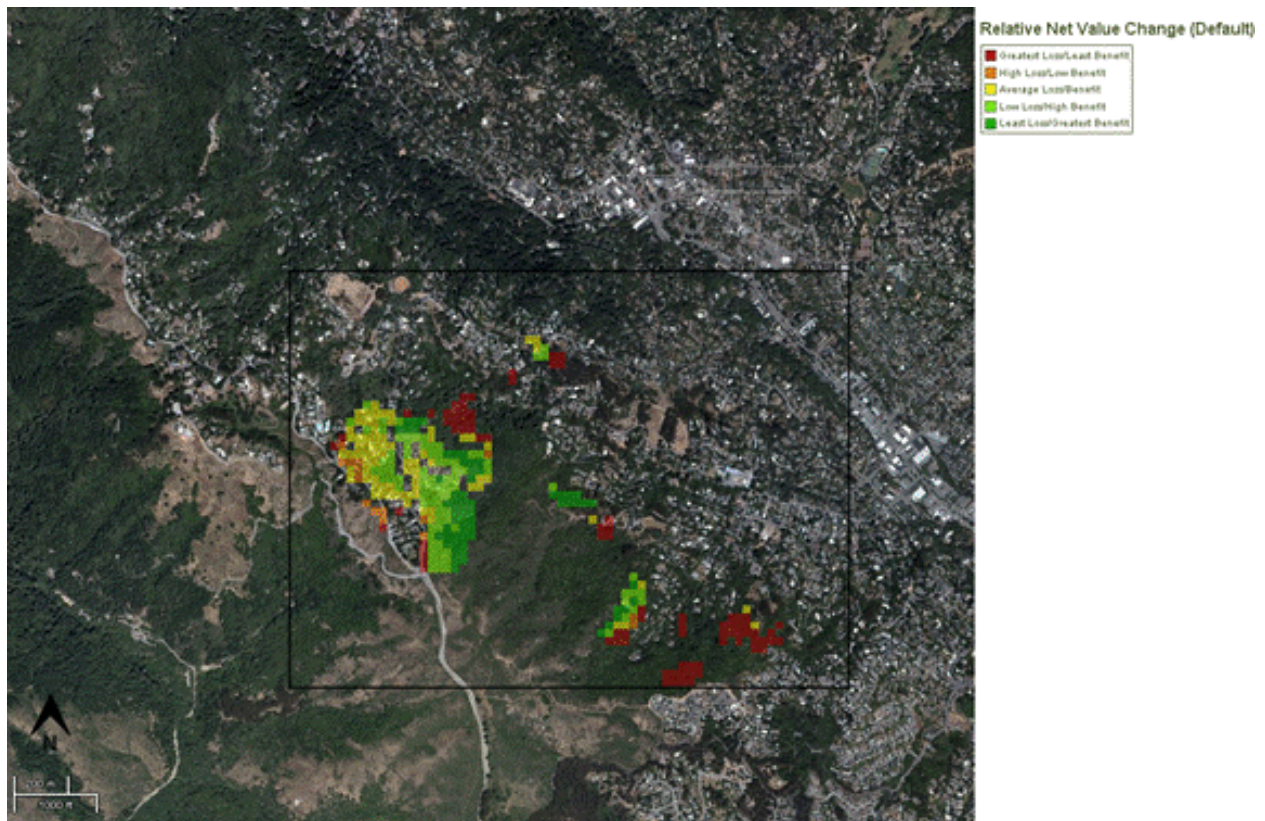
Under our second scenario, we assume the same type of weather pattern lead up but change the wind direction from N to E, or a variant thereof, to a turnaround day type of wind event. This would assume a fire starting late afternoon under a North or East wind blowing it on to offshore but by evening time, the on-shore push of marine layer would then refocus the fire back into homes from the GGNRA. It is also highly probable that a fire could have moved from outside of our area off of Mount Tamalpais during the prior burn period establishing itself West of Homestead Valley. Under this scenario, and onshore push would send a new head fire back up over the hill affecting Homestead Valley with an ember shower from Eucalyptus groves in the area at the top edges of the community and/or embers from structural ignitions on the western edge of the community near Sequoia Valley Road and Panoramic Highway.

On-shore Flow Flame Length Map – OCTOBER 2015



One can see here that contrasting Offshore flow vs. On-shore flow that flame lengths significantly decrease during onshore flow wildfire scenarios. However, without treatments and proper defensible space, the orange and red areas are still predicting flame lengths that will push the safe limits of firefighting ground resources except for dozers. Given the makeup of the area and proximity of residential streets, there is only small potential deployment areas for this kind of heavy equipment. Look at the map below which illustrates the “Pre-treatment” Risk Map. Using the IFTDSS modeling program under scenario 2, it shows where the highest losses will occur. You will notice in the next section that this map and the slope class and canopy height map start to line up showing our obvious areas for fuel modifications and treatment regimens.

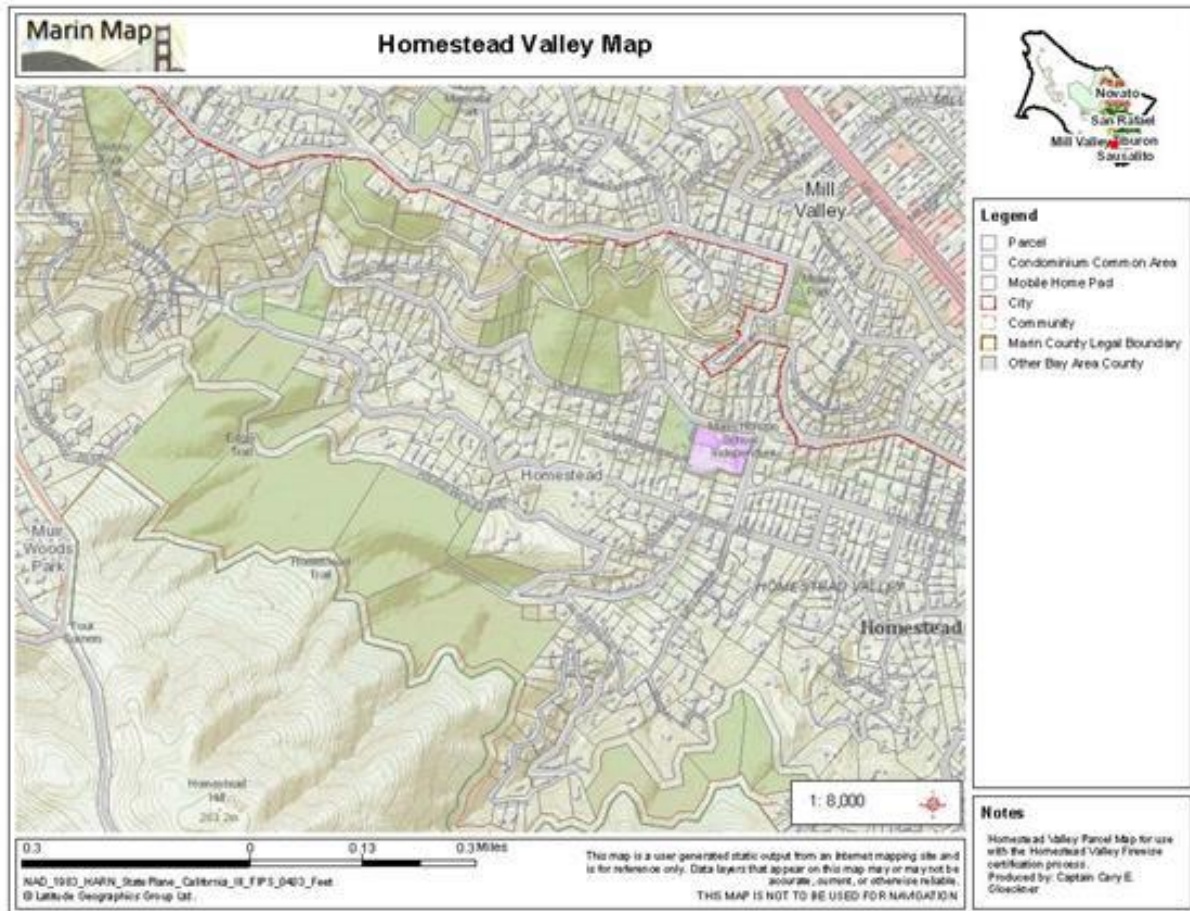
Pre-Treatment Risk Map – OCTOBER 2015



This scenario relies heavily on homeowners doing their part ahead of time to create defensible space, clean their roofs and gutters while also attending to larger scale projects like Eucalyptus removal, shaded fuel break construction and roadway encroachment/clearing projects. These sorts of man-made fuel breaks will serve to slow down or stop the forward advance of a downslope on-shore push type of wildfire.

4) Site Description – Homestead Valley:

In 1866 Samuel Throckmorton, owner of Rancho Sausalito, built a hunting lodge which he named “The Homestead” at what is now the corner of Ethel and Montford in a valley that later became known as Homestead Valley. In 1903, the Tamalpais Land & Water Company, which had become the owner of Rancho Sausalito, subdivided the valley into building sites. Homestead Valley today is an unincorporated community of about 1100 homes located between the Golden Gate National Recreation Area and the City of Mill Valley.

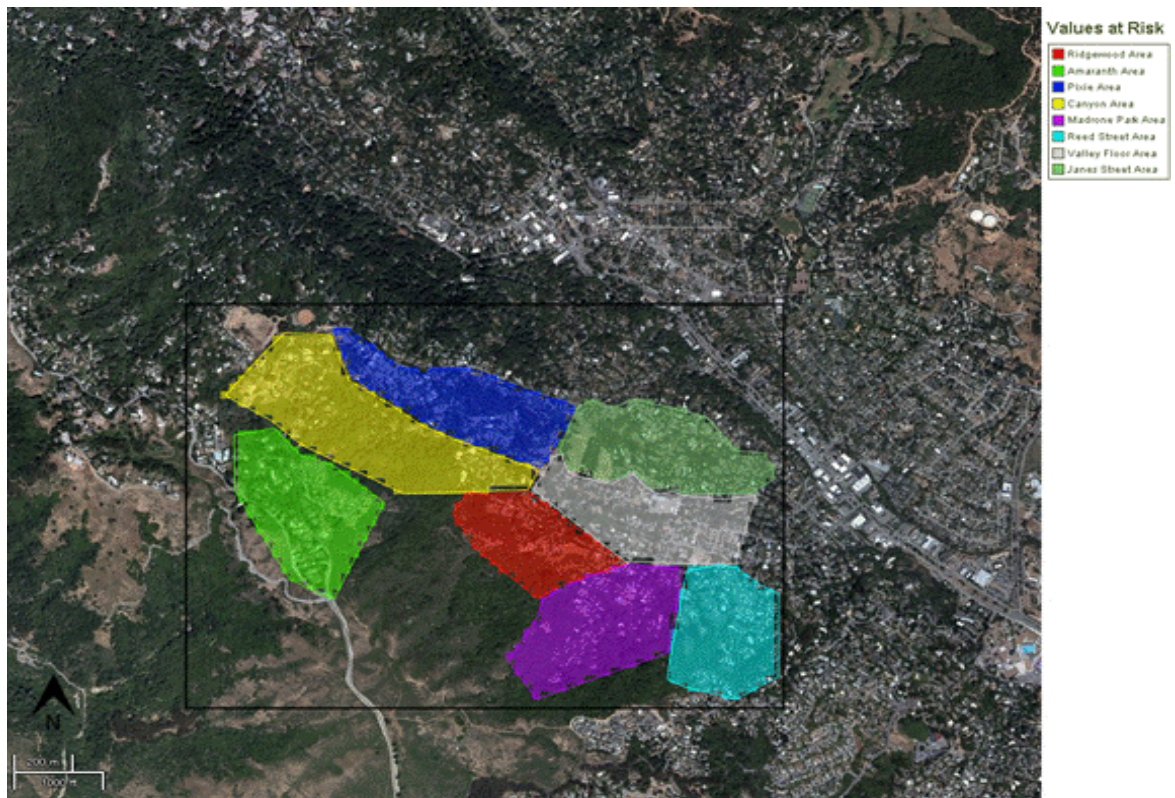


Homestead Valley residents have a strong history of preserving open space and supporting the Community Center. In 1967, Homestead Valley residents worked with the Marin County Board of Supervisors to create County Service Area (CSA) 14 to help finance the purchase of the open space and provide needed recreation facilities. In 1973, a \$600,000 bond issue was approved by HV voters to acquire open space and park properties and facilities. Added to that were funds from the sale of HVCA's Brown's Hall (on Miller Ave) and \$200,000 in matching funds from Marin County to purchase the site of the current Homestead Valley Community Center (formerly the Call residence) and over 85 acres of open space land, much of which borders land purchased by GGNRA.

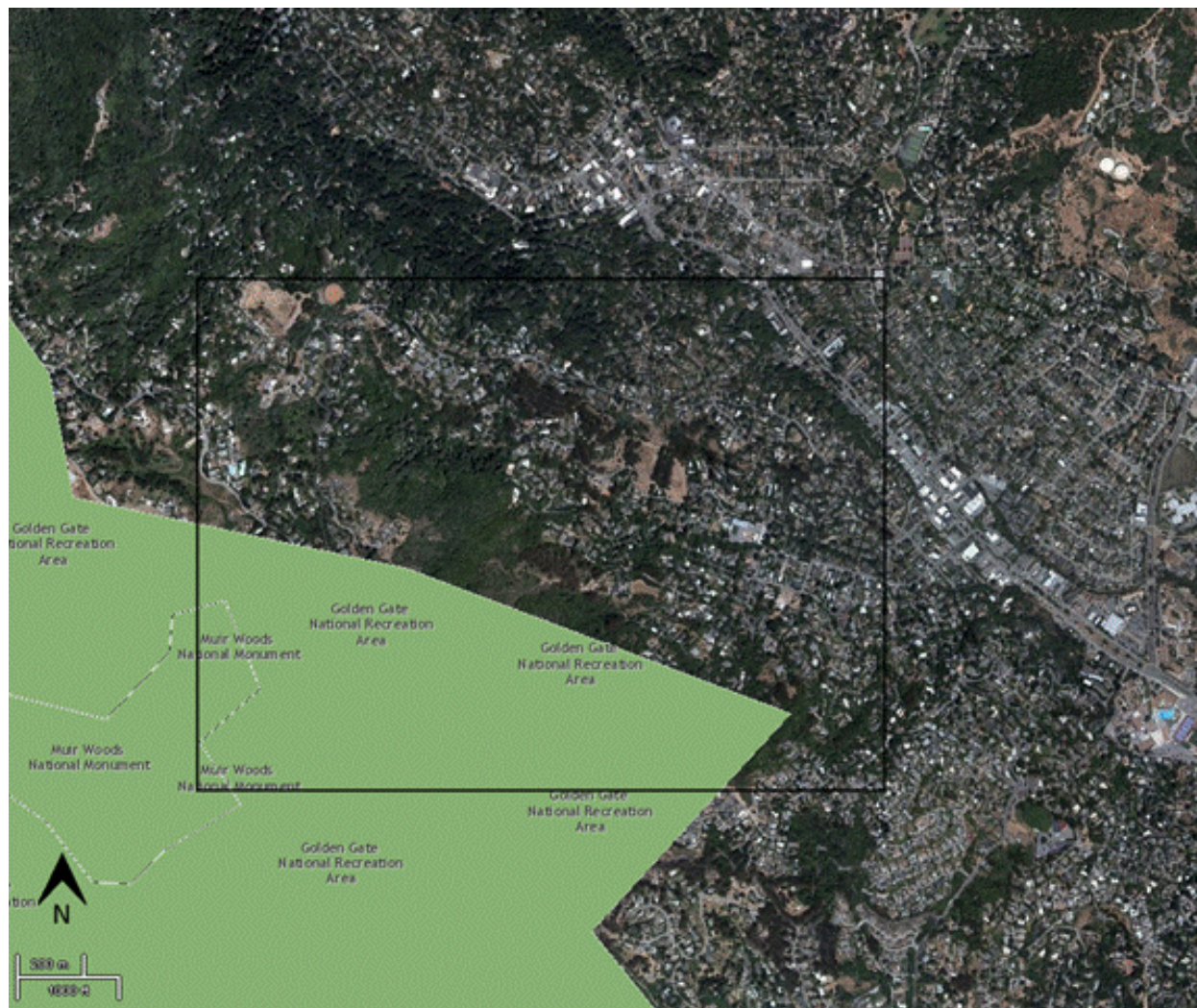
When the center was in need of significant repair and overgrowth was a concern in our open space, residents approved Measure A in 2009, a tax of \$125 per improved parcel for Community Center repair and renovation and open space fire hazard reduction. This special tax increases each year by 2% and does not have a sunset date. The total number of parcels included in this tax is 952. The meadow and open space are publicly owned in order to ensure its continued preservation for years to come.

Below is a map of "Values at Risk" that shows the 8 primary areas of concern outlined by a black box generally encapsulating the Homestead Valley Firewise area with a little overlap into other jurisdictions. Following this map is a map showing the overlap and proximity to untreated Federal lands. Most noticeably is the overlap and intermix of G.G.N.R.A. lands (Golden Gate National Recreation Area).

VALUES AT RISK MAP (8) – OCTOBER 2015



FEDERAL LANDS MAP – OCTOBER 2015



VEGETATION TYPES AND TOPOGRAPHY

Homestead Valley area has a large variety of highly diversified fuel models. During the Southern Marin Fire District Risk Assessment, completed by Captain Cary Gloeckner, the IFTDSS Program was utilized to assess the various fuel models which in turn created the maps contained herein illustrating the various fuel types. There are a few different fuel models that are used worldwide and some clarification as to which ones were used in Homestead Valley is needed.

As part of the process of predicting the potential behavior and effects of wildland fires, forest/fire scientists have used in specific fire behavior or fire effects models. Anderson (1982) proposed 13 fuel models; Scott and Burgan (2005) prop models (Rothermel, 1972 and 1983; Anderson, 1982; Scott and Burgan, 2005).

These collections of fuel properties are sometimes referred to more briefly as fuel models, though we prefer to refer to them as fire behavior fuel models because there are other kinds of fuel models that are used for other purposes.

The fire behavior fuel models are classified into four groups—grasses, brush (or shrubs); timber; and slash. The differences in fire behavior among these groups are basically related to the fuel load and its distribution among the fuel particle size classes (Anderson, 1982).

The fire behavior modules used in IFTDSS require these descriptions of fuel properties as input parameters, which are then used to calculate fire behavior potentials. In Captain Gloeckner's assessment, Scott and Burgan Fuel Models were used to predict fire behavior and to map Homestead Valley's fuel models. A map is included depicting these areas with a color-coded legend showing the following fuel bed types:

NB1: Urban/Developed – City and/or highly developed residential areas.

GR1 (101) & GR2 (102) - The primary carrier of fire in the GR fuel models is grass. Grass fuels can vary from heavily grazed grass stubble or sparse natural grass to dense grass more than 6 feet tall. Fire behavior varies from moderate spread rate and low flame length in the sparse grass to extreme spread rate and flame length in the tall grass models.

All GR fuel models are dynamic, meaning that their live herbaceous fuel load shifts from live to dead as a function of live herbaceous moisture content. The effect of live herbaceous moisture content on spread rate and intensity is strong.

GS1 (121) & GS2 (122) - The primary carrier of fire in the GS fuel models is grass and shrubs combined; both components are important in determining fire behavior. All GS fuel models are dynamic, meaning that their live herbaceous fuel load shifts from live to dead as a function of live herbaceous moisture content. The effect of live herbaceous moisture content on spread rate and intensity is strong and depends on the relative amount of grass and shrub load in the fuel model.

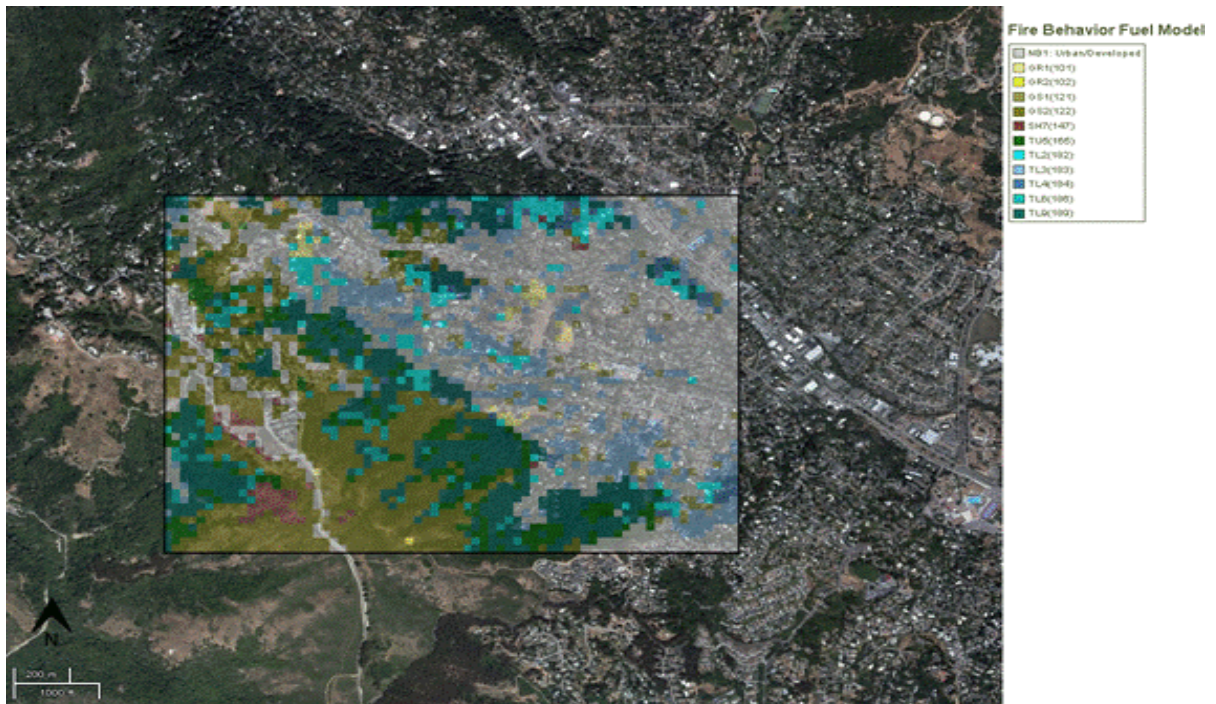
SH7 (147) - The primary carrier of fire in the SH fuel models is live and dead shrub twigs and foliage in combination with dead and down shrub litter. A small amount of herbaceous fuel may be present, especially in SH1 and SH9, which are dynamic models (their live herbaceous fuel load shifts from live to dead as a function of live herbaceous moisture content). The effect of live herbaceous moisture content on spread rate and flame length can be strong in those dynamic SH models.

TU5 (165) - The primary carrier of fire in the TU fuel models is forest litter in combination with herbaceous or shrub fuels. TU1 and TU3 contain live herbaceous load and are dynamic, meaning that their live herbaceous fuel load is allocated between live and dead as a function of live herbaceous moisture content. The effect of live herbaceous moisture content on spread rate and intensity is strong and depends on the relative amount of grass and shrub load in the fuel model.

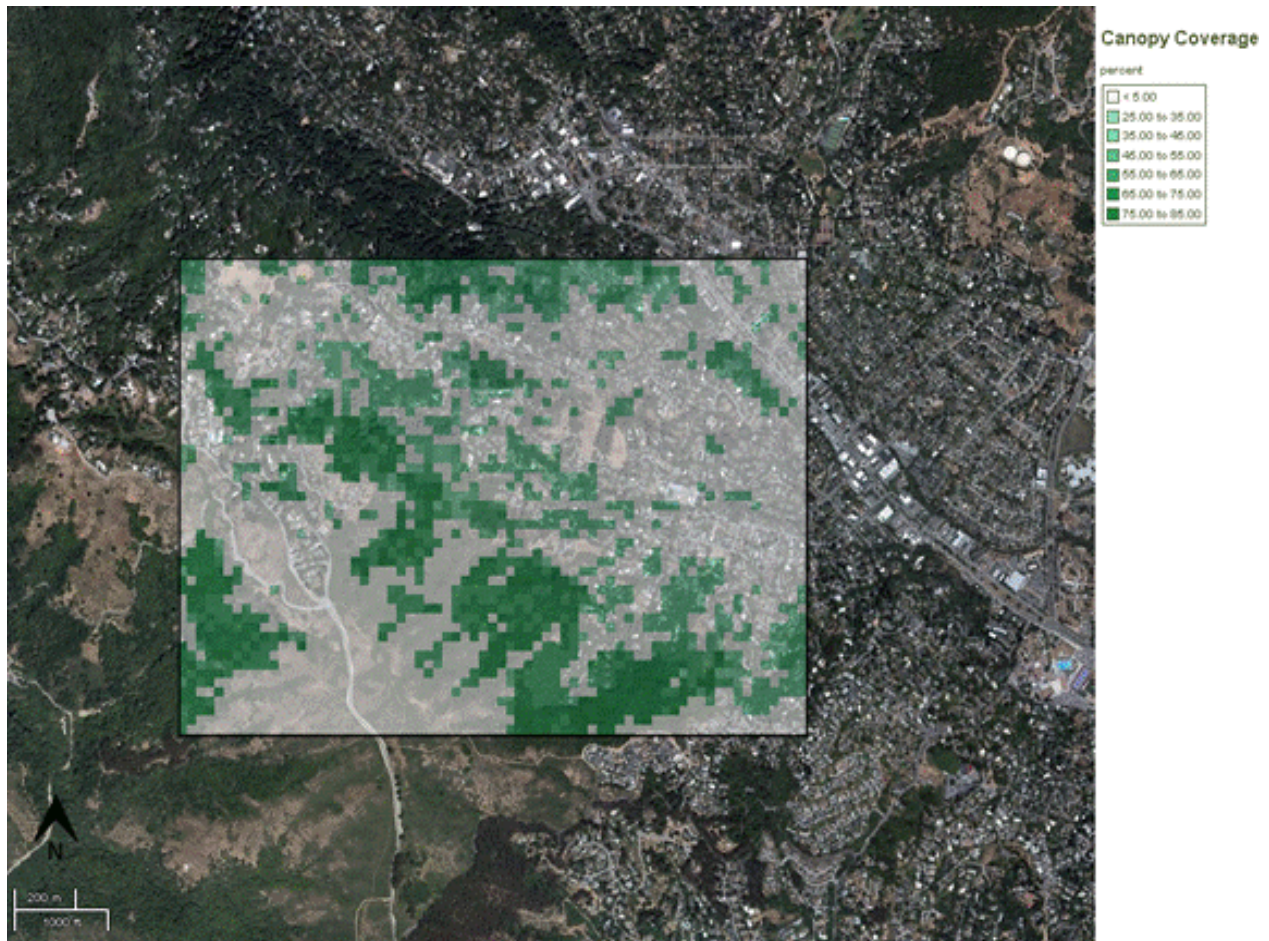
TL2 (182), TL3 (183), TL4 (184), TL6 (186), TL9 (189) - The primary carrier of fire in the TL fuel models is dead and down woody fuel. Live fuel, if present, has little effect on fire behavior.

Fire Behavior Fuel Model Classes HVFC, Canopy Coverage, Canopy Height maps are attached below illustrating our areas fuels and demonstrating a high density of Eucalyptus stands.

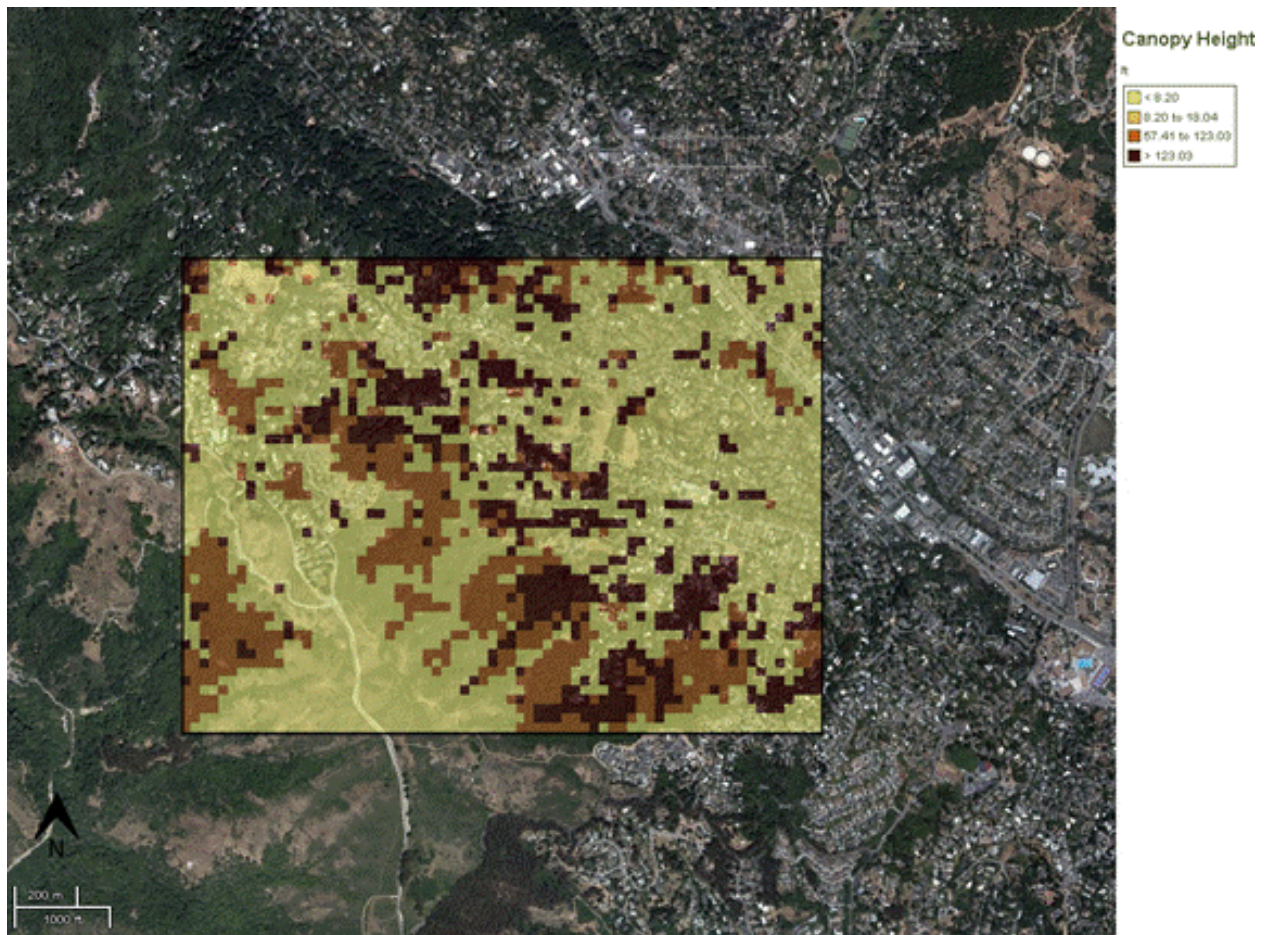
HOMESTEAD VALLEY FUEL MODEL CLASS MAP – OCTOBER 2015



HOMESTEAD VALLEY CANOPY COVERAGE MAP – OCTOBER 2015

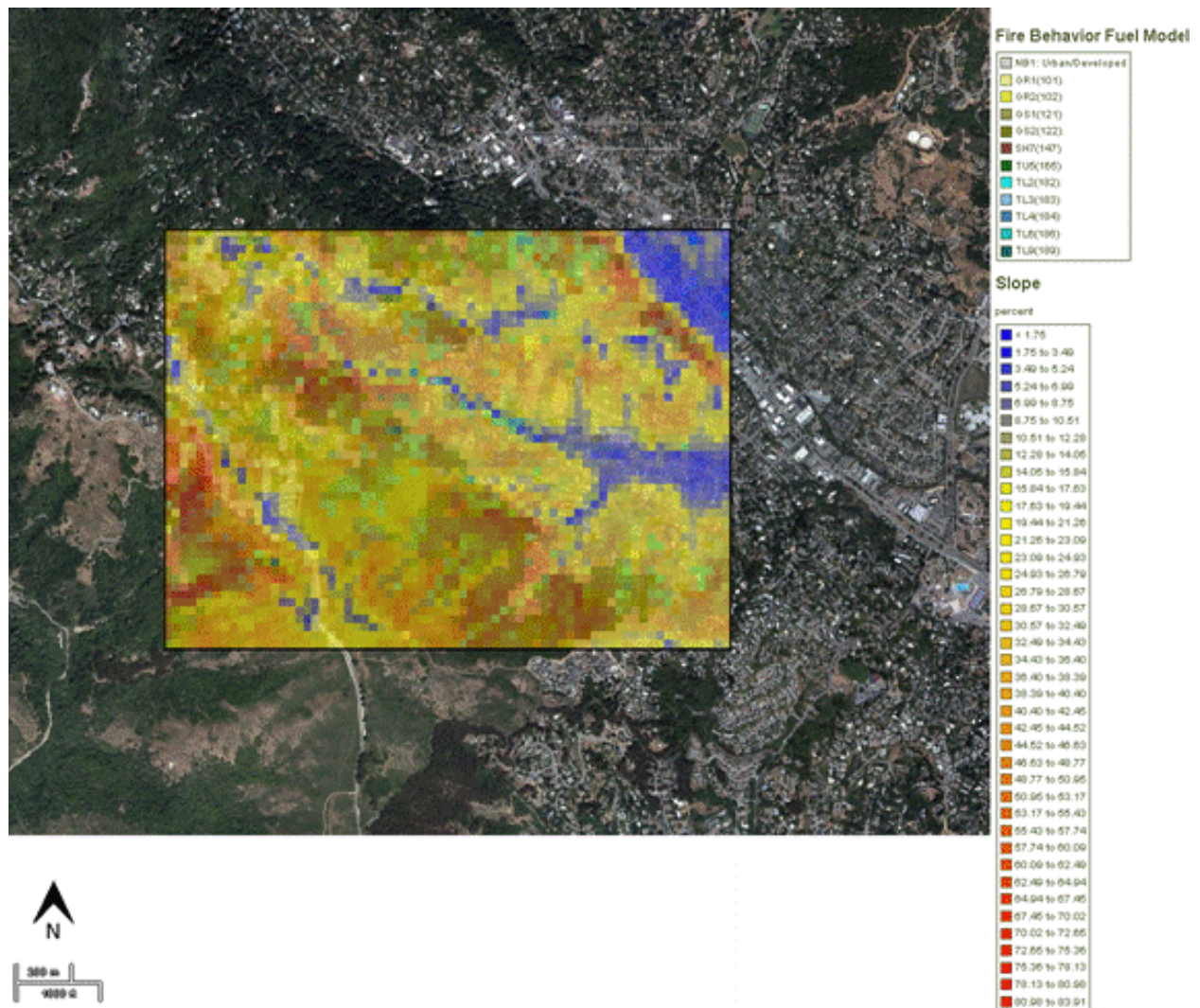


HOMESTEAD VALLEY CANOPY HEIGHT MAP – OCTOBER 2015



Another major factor in determining how a wildfire will move through a landscape depends a lot on the slope of the wildfire environment. Steeper slopes tend to expedite fire movement quicker than more gentle slopes. An analysis of slope was made using the IFTDSS program which generated the following slope class map seen below. What you will notice is pockets of extremely steep areas that one can also line up with our more dangerous vegetation types.

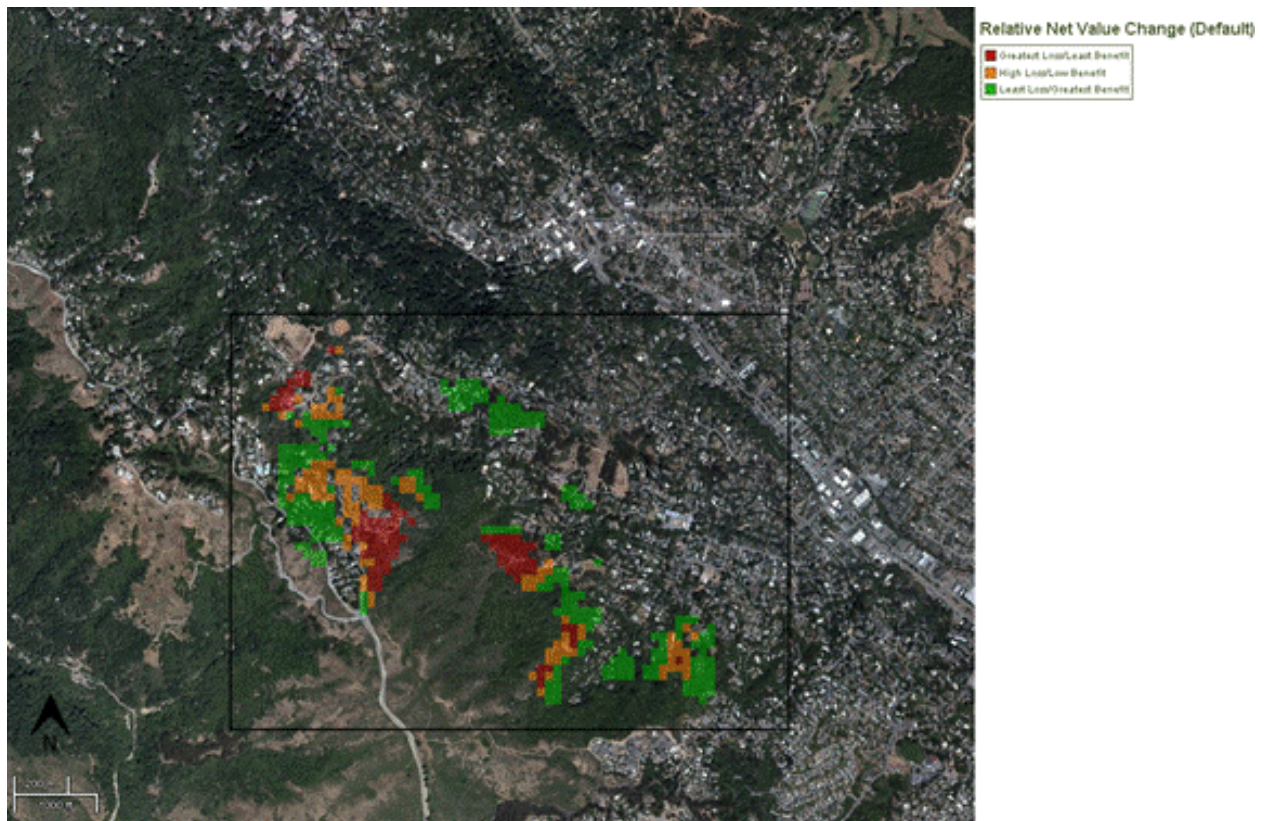
Slope Class Map – OCTOBER 2015



5) Assessment Process:

Captain Gloeckner has conducted multiple “Parcel-Level” NFPA 1144 assessments in the Pixie Trail area, Amaranth Area and other various locations throughout the Homestead Valley and coupled those with the “Worst Case Scenario” Fire Behavior Computer Modeling process using IFTDSS it has illustrated quite clearly, as seen above, that there exists about 5 high hazard wildfire loss areas.

High Loss Areas Map – OCTOBER 2015



Seen below is the Southern Marin Fire District version of the NFPA 1144 form used to rate an individual parcel for the threat of wildfire. This form has been used to evaluate and assess every home on Pixie Trail in Homestead Valley.

Site Address: _____

Assessment Element	Points	Score
One primary road in and out	10	
Roadway less than 12 feet wide	5	
Parked vehicles normally within 6 feet of the center of roadway	10	
Roadway encroached by vegetation or low hanging branches	10	
Dead-end roads greater than 200 feet	5	
Dominant Fuel Model and Defensible Space Treatment		
Light (grasses or irrigated and separated landscaping)	2	
Medium (light brush or un-cut grass under small trees)	5	
Heavy (dense brush or brush under hardwoods)	10	
Roof and/or gutters with leaves or needle cast build-up	10	
50 to 100 feet of defensible space from home	5	
No defensible space treatment or less than 50 feet	10	
Slope		
Gentle (0 to 30%)	2	
Moderate (30 to 45%)	5	
Steep (Greater than 45%)	10	
Additional Rating Factors		
1. Firewood pile(s) within 20 feet of the home	5	
2. Address improperly located, missing or nonreflective	5	
3. House closer than 100 ft. to untreated open space	5	
4. Blackberry, Juniper trees or shrubs, Pine needle cast, Pampas Grass w/in 50 feet of the house.	5	
5. Eucalyptus tree(s) within 1500 feet of house (1/4 mile or 500 yards)	10	
6. (Exterior) plastic collection barrels w/in 20 feet of the home	10	
7. Adjacent home(s) within 50 feet	10	
8. Area subject to strong winds	5	
Existing Building Construction		
Combustible Roof Materials (Anything other than A/C, Metal or Mineral)	15	
Vinyl siding	4	
T-111 siding or Old Growth Redwood Vertical Plank Siding	4	
Wood shake siding	15	
Combustible deck	10	
Single pane windows	5	
Water Source Availability (w/in 500 ft.) and Utilities (Gas and Electric)		
Orange Top or Higher GPM availability without drafting	2	
Red Top Hydrant and/or at draft level	10	
One utility above ground	5	
All utilities above ground	10	
TOTAL ASSESSMENT POINTS		
Extreme Hazard	156 to 200	
High Hazard	91 to 155	
Moderate Hazard	11 to 90	
Low Hazard	0 to 10	

6) Important Considerations:

The Firewise Communities/USA program seeks to create a sustainable balance that will allow communities to live safely while maintaining environmental harmony in a WUI setting. Homeowners already balance their decisions about fire protection measures against their desire for certain flammable components on their properties. It is important for them to understand the implications of the choices they are making. These choices directly relate to the ignitability of their home ignition zones during a wildfire.

During the assessment process, Captain Gloeckner noticed five recurring wildfire mitigation measures that need to be taken.

It was noticed that approximately 9 out of 10 homes in the Homestead Valley area had roofs and/or gutters that were full of combustible materials like leaves, needle cast and in some cases dead grass that had grown in the gutters. With ember cast being one of the key driving forces in conflagration type wildfire spread, the roofs and gutters must be free of needle cast in order to stand a chance of surviving.



The second mitigation issue noticed was that most main roads and driveways are choked with vegetation. This is a critical issue for a couple of reasons. Number one is that it reduces that available roadway width so that cars have very little room to pass. Number two is that it inhibits a rapid fire emergency response with engines due to the branches hitting the vehicle and windshield areas. This cause that apparatus to reduce speed so that it can navigate all of the encroaching vegetation and branches.

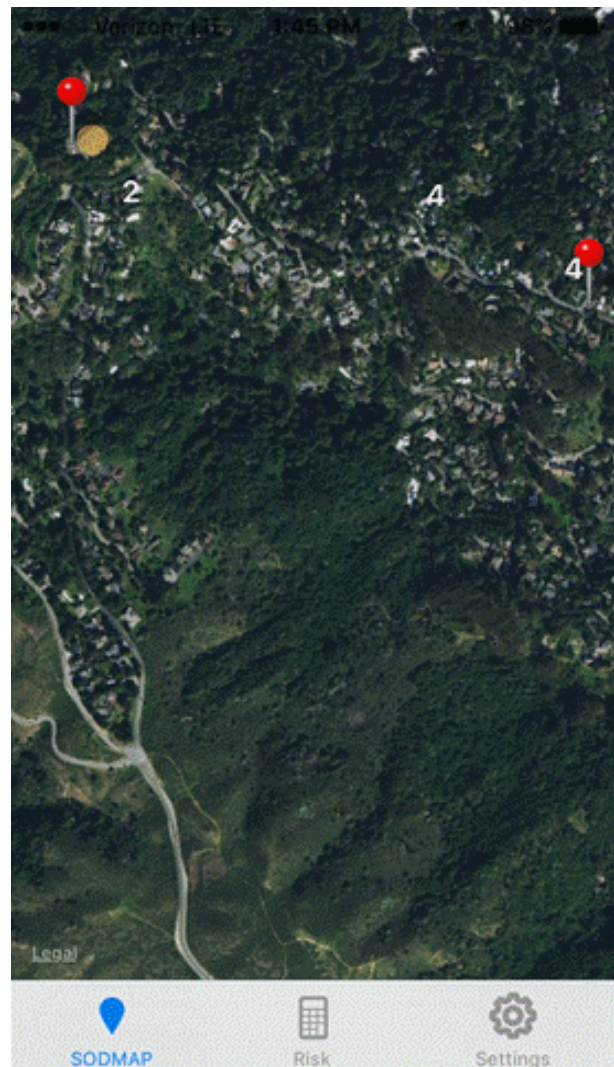


The third mitigation issue noticed was that most homes in the area do not have adequate defensible space out to the State mandated 100' either due to a lack of parcel size or knowledge of the necessity to do so. Another common reason in some of the areas that we forecast or modeled large flame lengths is that the homes are adjacent to untreated public lands (GGNRA). Please see the Federal Lands Map above.

A fourth mitigation issue noticed was the proximity to a high fire brand producing trees (Eucalyptus) which is well known to cause a very large fire brand issue during a wildfire. See the California Governor's Blue Ribbon Commission on the 1991 Tunnel Fire just 10 miles east of this location. As you can see in the picture here, there are numerous large untreated stands of Eucalyptus in and around the homes in the valley. This is a key area of concern and an area for great improvement in the coming years.



A fifth mitigation issue noticed was that amount of Sudden Oak Death syndrome noticed in the area. After years of drought, the areas Oak tree population is still seeing new breakouts of the tree killing disease. This can be seen on the image here that came from a University of Berkeley S.O.D. tree mapping program identifying key outbreaks along Montford Ave and at the head of the “Cowboy Trail” area. This disease creates a ton of extra dead and down material that during a wildfire will exacerbate the spread of fire and embers.



7A) Observations

1. Lack of defensible space around homes.
2. Roofs and gutters filled and/or covered with combustible debris.
3. Untreated Federal lands neighboring residential parcels.
4. Large untreated stands of Eucalyptus trees.
5. Roadways and driveways encroached by vegetation.
6. Lack of proper reflective address placards.
7. Open space lands overgrown.
8. Building siding consisting of wood shake shingles.
9. Clusters of Sudden Oak Death Trees.

7B) Recommendations:

1. Instruct homeowners through free public lectures on how to create a defensible space.
2. Community curbside chipper day events should take place to encourage brush removal and the creation of defensible space.
3. Media campaign targeting the cleaning of roofs and gutters of combustible debris.
4. Apply for grants to remove or clean up Eucalyptus stands.
5. Request Marin County Roads Department to cut back encroaching vegetation of public roadways. Encourage residents to do the same with their driveways.
6. Encourage the public to purchase code compliant reflective address signs through the Fire District.
7. Develop a vegetation management plan to address publicly held land such as the Homestead Valley Land Trust property, County Open spaces and Federal lands such as GGNRA.
8. Teach residents about the benefits of replacing old wood shingle siding with non-combustible siding or T-111 siding.
9. Apply for grants to remove dead and dying Sudden Oak Death trees.



8) Successful Firewise Modifications:

When adequately prepared, a house can likely withstand a wildfire without the intervention of the fire service. Further, a house and its surrounding community can be both Firewise and compatible with the area's ecosystem. The Firewise Communities/USA program is designed to enable communities to achieve a high level of protection against WUI fire loss even as a sustainable ecosystem balance is maintained.

A homeowner/community must focus attention on the home ignition zone and eliminate the fire's potential relationship with the house. This can be accomplished by disconnecting the house from high and/or low-intensity fire that could occur around it. The following photographs were taken in Homestead Valley Area and are examples of good Firewise practices.

1) Amaranth HOA defensible space implemented on the downslope side of the homes. This is a necessary practice in order to give firefighters a chance to get in and around the structures to protect them. Cutting back the thick brush and limbing up the trees in the area has modified the landscape so that an upslope wildfire with runs of crown fire will hit this transition area and lay down on the grassy slopes around the homes. These areas have been repeatedly cut and raked clean.



2) Pixie Trail road encroachment project was completed this summer (2015) by the residents. The first step was to use the NFPA 1144 form to rate their hazard which got them highly motivated to remove the encroaching vegetation. Residents pitched in and got chipper days so that the cost of removal was spread out amongst themselves. This in turn spurred on residents to go even farther by clearing out their backyards, finally establishing good defensible space.



3) Homestead Valley Land Trust has been continually spending money on thinning out the Eucalyptus stands on their open space property. This practice will help to keep wildfire on the ground and in the event that a wildfire makes it into the canopy, the thinning will keep the wildfire held at individual tree torching instead of perpetuating a running crown fire.



9) Next Steps:

After reviewing the contents of this assessment and its recommendations, the Homestead Valley Community Association (HVCA) Firewise Sub-Committee in cooperation with the Southern Marin (Captain Cary Gloeckner) will determine whether or not it wishes to continue seeking Firewise Communities/USA recognition. The Firewise Communities/USA representative will contact the HVCA Firewise Sub-Committee representative by [date] to receive its decision.

If the site assessment and recommendations are accepted and recognition will be sought, the HVCA Firewise Sub-Committee will create agreed-upon, area-specific solutions to the Firewise recommendations and create an action plan in cooperation with the Southern Marin Fire Department (Captain Cary Gloeckner – Firehouse Station #4).

Assuming the assessment area seeks to achieve national Firewise Communities/USA recognition status, it will integrate the following standards into its plan of action:

- Sponsor the HVCA Firewise Sub-Committee that maintains the Firewise Community program and status.
- Enlist a WUI specialist to complete an assessment and create a plan from which it identifies agreed-upon, achievable local solutions.
- Invest a minimum of \$2.00 annually per capita in its Firewise Communities/USA program. (Work done by municipal employees or volunteers, using municipal or other equipment, can be included, as can state/federal grants dedicated to that purpose.)
- Observe a Firewise Communities/USA Day each spring that is dedicated to a local Firewise project.
- Submit an annual report to Firewise Communities/USA. This report documents continuing participation in the program.

Homestead Valley residents are reminded to be conscious of keeping high-intensity fire more than 100 feet from their homes. It is important for them to avoid fire contact with their structures. This includes firebrands. The assessment team recommends the establishment of a ‘fire free zone’, allowing no fire to burn within ten feet of a house by removing fuels located there. It is a bad idea for fire to touch a house during a wildfire. Remember that, while wildfire cannot be eliminated from a property, it can be reduced in intensity.

Homeowners are reminded that street signs, addresses, road widths and fire hydrants do not keep a house from igniting. Proper attention to their home ignition zones does. They should identify the things that will ignite their homes and address those.

Weather is, of course, of great concern during wildfire season. At such time as fire weather is severe, homeowners should remember not to leave flammable items outside. This includes rattan doormats, flammable patio furniture, firewood stacked next to the house, or other flammables.